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Back to Basics: Basic Research Spillovers, Innovation Policy, and Growth

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1 Big picture

- **Question.** How do *basic* and *applied* research differ in the spillovers they generate, and what does that imply for innovation policy, growth, and welfare?
- **Core distinction.** Basic research can spill over across industries and can also raise the productivity of later applied innovations within an industry. Applied research is more targeted and mainly commercializes existing opportunities.
- **Institutional distinction.** The paper also separates *private* basic research from *public* basic research. Private basic research is more likely to be embodied quickly in products, while public basic research is more “ivory tower” and affects production with a delay.
- **Framework.** A Schumpeterian, multi-industry general equilibrium model with private firms, entrants, and public research labs. Firms may operate in multiple industries, and this breadth determines how much of basic-research spillovers they can internalize.
- **Data.** French firm-level data, 2000–2006, combining the annual R&D survey with firm-balance-sheet and ownership data, plus patent-citation information.
- **Empirical organizing fact.** Firms with broader multi-industry scope devote a larger share of research to basic research. This is the paper’s key reduced-form fact and the main identifying source for cross-industry spillovers.
- **Main quantitative result.** The decentralized equilibrium underinvests in basic research and overinvests in applied research. The dominant distortion is not “too little research overall,” but the *wrong composition* of research.
- **Headline policy result.** A uniform R&D subsidy is weak because it oversubsidizes applied research. The optimal targeted policy heavily subsidizes basic research (about 49%) and only mildly subsidizes applied research (about 11%).
- **Welfare result.** Relative to the constrained planner, the decentralized equilibrium loses about 4.6% in consumption-equivalent welfare. A uniform subsidy closes only a small fraction of that gap; a type-dependent subsidy closes much more.
- **Why this paper matters.** It provides a structural way to think about why “more R&D” is not enough: the type of research and the institution performing it matter for aggregate growth and policy design.

2 Data and measurement (key objects)

2.1 Core firm-level datasets

- **French Ministry of Research annual R&D Survey.** Main source for firm-level research expenditures, including a direct breakdown between *basic* and *applied* research.
- **LIFI and EAE datasets.** Used to recover firms’ activity across industries, both within the same firm and through ownership links inside business groups.
- **Balance-sheet data.** Used to measure profitability, firm size, growth, age, and entry/exit patterns.

- **NBER patent citation data.** Used to discipline innovation step sizes and the persistence of within-industry basic-research spillovers.
- **Sample size.** The final structural sample contains 13,708 firm-year observations for 2000–2006.

2.2 How the paper measures basic versus applied research

- **OECD/Frascati classification.** Basic research is “systematic study” without specific application in mind; applied research is directed toward a recognized practical need.
- **Key dependent variable in reduced-form evidence.**

$$\text{BRI}_{f,t} \equiv \frac{\text{basic research spending}_{f,t}}{\text{applied research spending}_{f,t}}. \quad (1)$$

- **Interpretation.** This is the paper’s empirical counterpart to the model’s research-composition choice. The slope of BRI against industry scope is informative about the cross-industry spillover parameter.

2.3 How the paper measures multi-industry scope

- **Number of industries.** The main scope variable is the number of distinct 1-digit SIC industries in which a firm is present, directly or through ownership links.
- **Why this matters.** In the model, a firm with activity in more industries is better able to appropriate the output of basic research because more spillovers land in places where it already has working knowledge.
- **Key descriptive fact.** The average innovative firm operates in about two 1-digit industries.

2.4 How the paper measures innovation quality and cooldown

- **Public vs private patents.** Patents assigned to public institutions proxy for basic-research-originating inventions; patents assigned to corporations proxy for commercially oriented inventions.
- **First-generation citations.** Used to pin down average innovation step sizes, because higher-step innovations receive more citations.
- **Second-generation citations.** Used to measure how long the “hot” basic-knowledge state lasts before cooling down.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production structure

The downstream final good is a Cobb–Douglas aggregate of industry outputs:

$$Z(t) = \prod_{i=1}^M Y_i(t)^{1/M}. \quad (2)$$

Each industry output is a CES aggregate of product lines:

$$Y_i = \left(\int_0^1 y_{ij}^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}}. \quad (3)$$

At the product-line level, the lead producer uses labor linearly:

$$y_{ij} = q_{ij} l_{ij}. \quad (4)$$

The industry productivity index is:

$$\bar{q}_i \equiv \left(\int_0^1 q_{ij}^{\epsilon-1} dj \right)^{1/(\epsilon-1)}. \quad (5)$$

- **Interpretation.** The model is a quality-ladder environment embedded in a multi-industry production network. A firm's product-line portfolio determines profits and innovative incentives.
- **Why $\bar{q}_i$ matters.** Innovation step sizes are scaled by industry productivity, so innovation is additive in normalized quality space and the firm-size distribution can remain stationary along the balanced growth path.

3.2 Innovation technologies: applied and basic research

For research type $s \in \{a, b\}$, the per-product-line research cost is:

$$h_s(s_i) = \xi_s s_i^{\nu_s}. \quad (6)$$

- **Interpretation.** Applied and basic research both use convex innovation technologies, but their cost curvature and scale can differ.
- **Key parameters.** ξ_s controls the level of cost, while $\nu_s > 1$ governs diminishing returns to research effort.
- **Fixed cost of basic research.** A firm with n product lines draws a fixed labor cost $n\phi$ for operating a basic-research lab. This creates an extensive margin for basic-research participation.

3.3 Basic research and cross-industry spillovers

A successful basic innovation in product line j of industry i raises the frontier quality by:

$$q_{ij}(t + \Delta t) = q_{ij}(t) + \eta \bar{q}_i(t). \quad (7)$$

The number of additional spillovers generated by that basic innovation is geometrically distributed:

$$F^n = (1 - s)s^n, \quad s = \frac{p}{1 + p}. \quad (8)$$

For a firm active in m industries, expected utilized spillovers are:

$$\text{Utilized spillovers}(m) = p \times \frac{m}{M}. \quad (9)$$

- **Interpretation.** p is the core cross-industry spillover parameter. It captures how broadly useful basic research is.
- **Economic logic.** Firms with broader industry scope can embody more of the discoveries triggered by basic research, so they have stronger private incentives to do it.
- **Main model prediction.** Basic-research intensity rises with multi-industry presence.

3.4 Within-industry spillovers: hot and cold product lines

Applied innovation depends on whether the latest basic knowledge in a product line is still active:

$$q_{ij}(t + \Delta t) = \begin{cases} q_{ij}(t) + \eta \bar{q}_i(t), & \text{if the line is hot,} \\ q_{ij}(t) + \lambda \bar{q}_i(t), & \text{if the line is cold.} \end{cases} \quad (10)$$

The hot/cold transition matrix is summarized by:

	hot	cold
hot	$1 - \zeta \Delta t$	$\zeta \Delta t$
cold	$\tau_b \Delta t$	$1 - \tau_b \Delta t$

- **Interpretation.** Basic research creates a temporary local knowledge frontier. Applied research commercializes it.
- **Key parameter.** ζ is the cooldown rate. A lower ζ means basic knowledge remains useful for follow-on innovation for longer.
- **Deep intuition.** Basic research is valuable not only because it creates products directly, but because it raises the expected step size of later applied innovation.

3.5 Public basic research

Each public lab generates basic innovation flow u subject to the same basic-research technology, financed by a transfer $\bar{R}$:

$$\bar{R} = w[h_b(u) + \bar{\phi}]. \quad (11)$$

- **Interpretation.** Government can directly shift the stock of basic knowledge by financing public labs.
- **Important asymmetry.** Public basic innovations affect later private applied innovation, but are not immediately embodied into products. This is the “ivory tower” feature.

3.6 Firm value and research incentives

Monopoly profit from a product line with normalized productivity $\hat{q}$ is:

$$\pi(\hat{q}) = \hat{q}^{(\epsilon-1)/\epsilon} \frac{Z}{M}. \quad (12)$$

Firm value decomposes additively:

$$V(\hat{q}, m) = \frac{Z}{M} \left[\sum_{\hat{q} \in \hat{Q}} V(\hat{q}) + nV_m \right]. \quad (13)$$

The key option-value term satisfies:

$$(r - g)V_m = \max_{a,b} \left\{ -\tilde{w} \left[(1 - \psi_a)h_a(a) + (1 - \psi_b)(h_b(b) + \mathbf{1}(b > 0)\phi) \right] \right. \\ \left. + a[\alpha V^H + (1 - \alpha)V^C + V_m] \right. \\ \left. + b \left(1 + p \frac{m}{M} \right) (V^H + V_m) + x \left(1 - \frac{m}{M} \right) (V_{m+1} - V_m) - \tau V_m + \kappa \mathbb{E}_{\hat{q}} V(\hat{q}_t) \right\}. \quad (14)$$

The optimal basic-research intensity per product line is:

$$b_m = \left[\frac{(1 + p \frac{m}{M}) (V^H + V_m)}{(1 - \psi_b) \nu_b \xi_b \tilde{w}} \right]^{1/(\nu_b - 1)}. \quad (15)$$

The optimal applied-research intensity is:

$$a = \left[\frac{\alpha V^H + (1 - \alpha)V^C}{(1 - \psi_a) \nu_a \xi_a \tilde{w}} \right]^{1/(\nu_a - 1)}. \quad (16)$$

- **Interpretation of b_m .** Basic research rises with industry scope because broader firms internalize more spillovers.
- **Interpretation of a .** Applied research is more attractive when the economy has more hot product lines, so basic and applied research are strategic complements.

3.7 Embodied vs disembodied basic research

The paper decomposes aggregate basic innovation into embodied and disembodied components:

$$\tau_b^e \equiv \sum_{m=1}^M \mu_m P_m (1 + \rho_m) b_m, \quad (17)$$

$$\tau_b^d \equiv \sum_{m=1}^M \mu_m P_m (p - \rho_m) b_m + (1 + p)u, \quad (18)$$

$$\tau_b \equiv \tau_b^e + \tau_b^d. \quad (19)$$

The steady-state hot share is:

$$\alpha = \frac{\tau_b}{\zeta + \tau_b}. \quad (20)$$

- **Interpretation.** This is the bridge between public policy and private-sector productivity. More public or private basic research raises α , which then raises the returns to applied research.

3.8 Aggregate innovation, labor allocation, and growth

Aggregate applied innovation is:

$$\tau_a = \sum_{m=1}^M \mu_m a_m + E a_e, \quad (21)$$

and creative destruction is:

$$\tau = \tau_a + \tau_b^e. \quad (22)$$

The labor-market-clearing condition is:

$$1 = \tilde{w}^{-(1-\epsilon)/\epsilon} + \sum_m \mu_m P_m [h_b(b_m) + \bar{\phi}_m] + U[h_b(u) + \bar{\phi}] + h_a(a) + E h_a(a_e). \quad (23)$$

Aggregate output is:

$$Z = \bar{q} L_p / M. \quad (24)$$

Balanced-growth-path growth is:

$$g = \frac{\tau_a [\alpha \mathbb{E}_{\hat{q}}^H (\hat{q} + \eta)^{\epsilon-1} + (1 - \alpha) \mathbb{E}_{\hat{q}}^C (\hat{q} + \lambda)^{\epsilon-1} - 1] + \tau_b^e [\mathbb{E}_{\hat{q}} (\hat{q} + \eta)^{\epsilon-1} - 1]}{\epsilon - 1}. \quad (25)$$

- **Interpretation.** Growth comes from both embodied basic innovation and applied innovation, but applied innovation is more productive when there is a larger stock of hot lines.
- **Policy implication.** Public basic research can raise growth indirectly by making private applied innovation more effective.

3.9 Welfare and the constrained planner

Balanced-growth welfare is:

$$W(C_0, g) = \int_0^\infty e^{-\delta t} \frac{(C_0 e^{gt})^{1-\gamma} - 1}{1-\gamma} dt = \frac{1}{1-\gamma} \left(\frac{C_0^{1-\gamma}}{\delta - (1-\gamma)g} - \frac{1}{\delta} \right). \quad (26)$$

The planner chooses common applied and basic innovation intensities and the threshold fixed cost for basic labs:

$$\max_{a, b, \hat{\phi}} (1 - \gamma) \log(1 - R) - \log(\delta - (1 - \gamma)g) \quad (27)$$

subject to

$$P = P[\phi < \hat{\phi}], \quad \tau_a = (1 + E)a, \quad \tau_b = (P + U)(1 + p)b, \quad (28)$$

$$R = (1 + E)h_a(a) + (P + U)h_b(b) + P\mathbb{E}[\phi \mid \phi < \hat{\phi}] + U\bar{\phi}, \quad (29)$$

with $g = g(\tau_a, \tau_b)$.

- **Interpretation.** The planner internalizes all basic-research spillovers, but still takes monopoly distortions on the production side as given.
- **Key lesson.** The big inefficiency is not “too little research overall” but misallocation across basic and applied research types.

3.10 Reduced-form and structural estimation equations

The main reduced-form relationship behind cross-industry identification is:

$$\text{BRI}_{f,t} = \beta \log(\text{number of industries}_{f,t}) + \Gamma X_{f,t} + \text{FE} + \varepsilon_{f,t}, \quad (30)$$

estimated by Tobit because many firms report zero basic-research spending.

The historical-IV version instruments scope with past state participation:

$$\log(\text{industries}_{f,t}) = \pi \text{StateParticipation1985}_f + \Pi X_{f,t} + \text{FE} + u_{f,t}. \quad (31)$$

The structural parameters are estimated by simulated method of moments:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} [\Lambda(\theta) - \Lambda(y)] W [\Lambda(\theta) - \Lambda(y)]. \quad (32)$$

- **Interpretation.** The reduced form validates the spillover mechanism directly in the data; the SMM then disciplines the full general-equilibrium model with moments on research, firm dynamics, and patent citations.

4 Identification and instruments (what makes the estimates credible)

4.1 Cross-industry spillovers from firms' research composition

- **Key idea.** If basic research creates spillovers across industries, then firms operating in more industries should devote more resources to basic research because they can appropriate more of the benefits.
- **Empirical support.** Basic-research intensity rises with the number of 1-digit SIC industries. The estimated marginal effect is about 0.024–0.032 in Tobit specifications, even after adding industry controls, research-area controls, and pre-sample patent controls.
- **Structural mapping.** The slope of basic-research intensity against industry scope identifies p , the cross-industry spillover parameter.
- **Extensive-margin information.** The probability of doing any basic research by scope class helps identify the mean and variance of the fixed cost of running basic-research labs, $\bar{\phi}$ and σ .

4.2 Instrumental-variable evidence

- **Instrument.** Historical state participation in 1985–87, during the French nationalization and conglomeration episode.
- **Why it helps.** State-led conglomeration changed firms' industrial scope for reasons plausibly outside their later research-composition choices.
- **First stage.** State participation in the 1980s predicts about 1.5 more industry links in 2000–06.
- **Second stage.** The instrumented effect of industry scope on basic-research intensity is positive and larger than the OLS estimate, which supports the spillover interpretation.
- **Main credibility argument.** The relationship remains after controlling for current state ownership, public R&D subsidies, and university outsourcing links.

4.3 Within-industry spillovers and cooldown from citations

- **Step-size identification.** Public patents receive more first-generation citations than private patents, consistent with larger basic-research step sizes.
- **Cooldown identification.** Second-generation citations to patents that build on public patents are higher for a while, but this advantage fades after roughly eight years. That decay disciplines the cooldown rate ζ .
- **Model-based moment.** The average time until a public innovation becomes observationally indistinguishable from a private one is

$$\Lambda(24) = \frac{1}{\zeta} \left(\frac{\tau_a}{\tau} \right). \quad (33)$$

- **Interpretation.** Basic knowledge matters temporarily; it is not a permanent productivity shifter for a line.

4.4 Other parameters

- **Innovation step sizes η, λ .** Identified mainly from citation distributions plus aggregate growth and firm profitability.
- **Entry mass E and buyout rate ς .** Identified from the mean and dispersion of firms' industry scope.
- **Research cost parameters $(\xi_a, \xi_b, \nu_a, \nu_b)$.** Identified from the overall level of research intensity, firm dynamics, and the model's ability to match observed research composition.

5 Tables and figures

5.1 Figures 1–3 and Table 1: model mechanics

$$\mathbf{q}(t) \equiv \begin{matrix} & i=1 & i=2 & i=3 & i=4 \\ \left\{ \begin{array}{l} q_{1,j_1}(t) \\ q_{1,j_2}(t) \\ q_{1,j_3}(t) \end{array} \right. & & & q_{3,j_1}(t) & q_{4,j_1}(t) \\ & & & q_{3,j_2}(t) & \end{matrix} \right\}.$$

FIGURE 1
Payoff-relevant state $\mathbf{q}$

$$\mathbf{q}(t) \equiv \begin{matrix} & i=1 & i=2 & i=3 & i=4 \\ \left\{ \begin{array}{l} q_{1,j_1}(t) \\ q_{1,j_2}(t) \\ q_{1,j_3}(t) \\ q_{1,j_4}(t) + \eta \bar{q}_1(t) \end{array} \right. & & & q_{3,j_1}(t) & q_{4,j_1}(t) \\ & & & q_{3,j_2}(t) & \end{matrix} \right\}.$$

FIGURE 2
Impact of basic innovation in industry $i=1$

$$\mathbf{q}(t) \equiv \begin{matrix} & i=1 & i=2 & i=3 & i=4 \\ \left\{ \begin{array}{l} q_{1,j_1}(t) \\ q_{1,j_2}(t) \\ q_{1,j_3}(t) \\ q_{1,j_4}(t) + \eta \bar{q}_1(t) \\ q_{1,j_5}(t) + \eta \bar{q}_1(t) \end{array} \right. & & & q_{3,j_1}(t) & q_{4,j_1}(t) \\ & & & q_{3,j_2}(t) & \\ & & & q_{3,j_3}(t) + \eta \bar{q}_3(t) & \end{matrix} \right\}.$$

FIGURE 3
Cross-industry spillover into industry $i=3$

TABLE 1 Transition matrix for within-industry spillovers		
	hot	cold
hot	$1 - \zeta \Delta t$	$\zeta \Delta t$
cold	$\tau_b \Delta t$	$1 - \tau_b \Delta t$

Read:

- Figures 1–3 are not “results” figures; they are teaching figures. They show the state vector of a multi-industry firm, the effect of a successful basic innovation within an industry, and how the same basic innovation can create additional product lines in other industries the firm already knows.
- Table 1 translates the basic idea of “hot” and “cold” product lines into a simple transition matrix. This is the whole within-industry spillover mechanism in one place.
- These exhibits are essential because later policy results all work through the distinction between embodied basic innovation, disembodied basic knowledge, and the hot-share α .

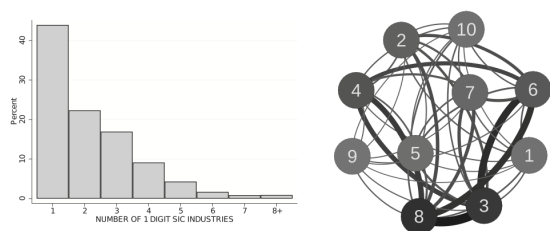


FIGURE 4
Distribution and links of firms across industries

The figure uses 13,708 firm-year observations from the pooled data for the period 2000–06. The left panel plots the share of firms as a function of the number of 1-Digit SIC industries, which is the number of distinct SIC codes in which a firm is present. The right panel plots the pairwise presence of firms across industries through nodes and lines. Node colour is assigned in terms of total number of firms in the industry (lighter to darker). Thickness of lines between industries is assigned in terms of the total number of links between each pair. Activity classification: 1 “Agriculture,” 2 “Food and Textile Industries,” 3 “Manufacture of chemicals, metals, and machinery,” 4 “Manufacture of electrical and transport equipment,” 5 “Construction and Utilities,” 6 “Wholesale and retail trade,” 7 “Transport, Communication, and Financial Activities,” 8 “Professional, scientific, and technical activities,” 9 “Education and Human Health Activities,” 10 “Arts, entertainment, and others.”

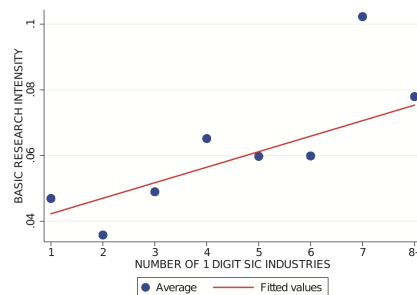


FIGURE 5
Basic research intensity across industries

The figure plots *Basic Research Intensity* as a function of *Number of 1-Digit SIC Industries*. For each firm-year observation *Basic Research Intensity* is defined as the ratio of total firm investment in basic research divided by total firm investment in applied research. *Average* is the average basic research intensity for firms conditional on the number of their 1-Digit SIC industries and after partialling out for firm size, while the red line represents a linear fit of the firm-year observations.

	(1)	(2)	(3)	(4)
Log no. of industries	0.032*** (0.006)	0.025*** (0.008)	0.024*** (0.008)	0.024*** (0.008)
Log employment	0.003** (0.001)	0.002 (0.001)	0.002 (0.001)	−0.000 (0.002)
Year and organization FE	Yes	Yes	Yes	Yes
Industry FE	No	Yes	Yes	Yes
Areas of research shares	No	No	Yes	Yes
Patent stock and no patent FE	No	No	No	Yes
N	13,708	13,708	13,707	13,708

Notes: Pooled data for the period 2000–06. Estimates are obtained using Tobit models and relate to the marginal effect of the variables at the sample mean. *Basic Research Intensity* is defined as the ratio of total firm investment in basic research divided by total firm investment in applied research. *Log no. of industries* is the number of distinct SIC codes in which a firm is present. *Log employment* is the total employment of firms. *Share in software/biotech/materials* is the share of R&D investment in the respective areas of research with share of environment research omitted. *Log patent stock* measures the number of patents obtained by firms in the pre-sample period (1993–2000). *Industry FE* denote fixed effects for 1-digit SIC activities. *Year FE* denotes year fixed effects, and *Organization FE* denotes whether the firm operates its activity as a conglomerate or as a business group. See the [Supplementary Appendix](#) for the definition of variables. Robust standard errors clustered at the firm level are in parentheses. One star denotes significance at the 10% level; two stars denote significance at the 5% level; and three stars denote significance at the 1% level.

	1st stage			2nd stage		
	(1)	(2)	(3)	(4)	(5)	(6)
State participation in 1985	0.293*** (0.056)	0.288*** (0.057)	0.264*** (0.056)			
Log no. of industries				0.085*** (0.018)	0.079*** (0.021)	0.070** (0.028)
Log employment	0.120*** (0.003)	0.120*** (0.004)	0.117*** (0.004)	−0.007** (0.003)	−0.006 (0.004)	−0.006 (0.004)
State participation	−0.057 (0.083)	−0.057 (0.083)	−0.049 (0.082)	−0.003 (0.011)	−0.005 (0.012)	−0.002 (0.013)
Public R&D subsidies		0.013* (0.008)			0.010*** (0.004)	
Outsourcing to Univ.			0.067*** (0.012)			0.019** (0.010)
Year and organization FE	Yes	Yes	Yes	Yes	Yes	Yes
Areas of research shares	Yes	Yes	Yes	Yes	Yes	Yes
N	13,707	13,707	13,707	13,707	13,707	13,707

Notes: Pooled data for the period 2000–06. Estimates are obtained using Tobit models and relate to the marginal effect of the variables at the sample mean. *Basic research intensity* is defined as the ratio of total firm investment in basic research divided by total firm investment in applied research. *Log no. of industries* is the number of distinct SIC codes in which a firm is present. *State participation in 1985* is a binary variable equal to 1 if the French state was a shareholder of the firm in 1985. *State participation* is a binary variable equal to 1 if the French state is currently a shareholder of the firm. *Public R&D subsidies* is a binary variable equal to 1 if the firm obtained public R&D subsidies. *University collaboration* is a binary variable equal to 1 if the firm collaborates/outsources some of its R&D to the public system. *Log employment* is the total employment of firms. *Share in software/biotech/materials* is the share of R&D investment in the respective areas of research. *Industry FE* denote fixed effects for 1-digit SIC activities. *Year FE* denotes year fixed effects, and *Organization FE* denotes whether the firm operates its activity as a conglomerate or as a business group. See the [Supplementary Appendix](#) for the definition of variables. Robust standard errors clustered at the firm level are in parentheses. One star denotes significance at the 10% level; two stars denote significance at the 5% level; and three stars denote significance at the 1% level.

5.2 Figure 4 and Figure 5: descriptive evidence on scope and basic research

Read:

- Figure 4 shows that innovative firms are often multi-industry firms; the average is around two industries, and cross-industry links are concentrated around manufacturing and scientific activities.
- Figure 5 is the key descriptive figure: after partialling out firm size, the ratio of basic to applied research rises with the number of industries.

- This figure is the reduced-form visual counterpart to the model’s central mechanism: broader scope raises the appropriability of basic research.

5.3 Tables 2 and 3: reduced-form and IV evidence

Read:

- Table 2 shows the positive relationship between industry scope and basic-research intensity is robust to firm-size controls, industry fixed effects, research-area controls, and patent-history controls.
- Table 3 addresses reverse causality using historical state participation as an instrument for multi-industry scope. The instrumented effect remains positive and is even larger.
- The combination of Tables 2 and 3 gives the reader confidence that the multi-industry/basic-research link is not a simple compositional artifact.

5.4 Table 4 and Figure 6: citation evidence

TABLE 4
Patent citations across public and private patents

	Mean	Panel A: 1st generation patents					Min	Max	N
		25th	Median	75th	St. Dev.				
Private citations	5.88	1.00	4.00	7.00	8.01	0.00	179.00	28247.00	
Public citations	7.01	2.00	4.00	8.00	11.86	0.00	351.00	2806.00	

Age cohort	Panel B: 2nd generation patents									
	1	2	3	4	5	6	7	8	9	10
Mean difference	0.3** (0.15)	0.3** (0.15)	0.62*** (0.17)	0.28** (0.14)	0.41** (0.18)	0.23 (0.17)	0.71*** (0.25)	0.08 (0.16)	0.39 (0.25)	.14 (0.24)
Adjusted	0.3** (0.15)	0.23* (0.14)	0.55*** (0.17)	0.26* (0.17)	0.46** (0.17)	0.18 (0.17)	0.58** (0.24)	0.06 (0.15)	0.47* (0.26)	.22 (0.28)

Notes: Citation patterns of patents granted by the USPTO to French private and public depositors. Panel A computes direct 1st generation citations across public and private patents. Panel B reports the average difference in citations of 2nd generation patents for the period 1975–90. Adjusted citations are conditioned on being within the same 1-digit industry of use, as indicated by the concordance table in Silverman (2004). Two sample t-tests with unequal variances were used. One star denotes significance at the 10% level; two stars denote significance at the 5% level; and three stars denote significance at the 1% level.

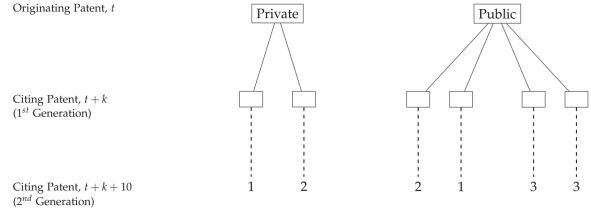


FIGURE 6
Construction of citation information

Read:

- Panel A of Table 4 shows public patents receive more direct citations than private patents, supporting the idea that basic research has larger step sizes.
- Panel B shows that follow-on patents citing public-origin patents remain more highly cited for several years, but the gap shrinks with age. This is what identifies cooldown.
- Figure 6 clarifies the logic of first-generation versus second-generation citations. It is crucial for understanding why the paper can separately identify step sizes and spillover persistence.

5.5 Figures 7–10: structural identification

Read:

- Figure 7 gives the citation model that links innovation step sizes to observed citation counts.
- Figures 8 and 9 show the model reproduces the distributions of private and public patent citations reasonably well.
- Figure 10 is especially important: changing the spillover parameter p rotates the model’s basic-research-intensity profile against industry scope. This makes the identification of p very transparent.



Basic innovations are represented by "B", while applied innovations are represented by "A". Each arrow represents the possibility of a citation. An actual citation will occur with probability ηx for basic innovations and λx for applied innovations.

FIGURE 7

Citation model diagram

Basic innovations are represented by "B", while applied innovations are represented by "A". Each arrow represents the possibility of a citation. An actual citation will occur with probability ηx for basic innovations and λx for applied innovations.

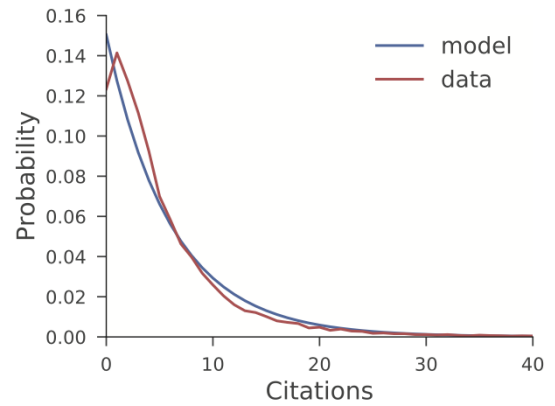


FIGURE 8

Private citation distributions

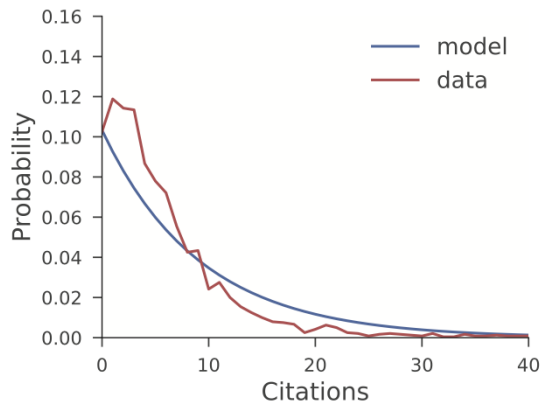


FIGURE 9

Public citation distributions

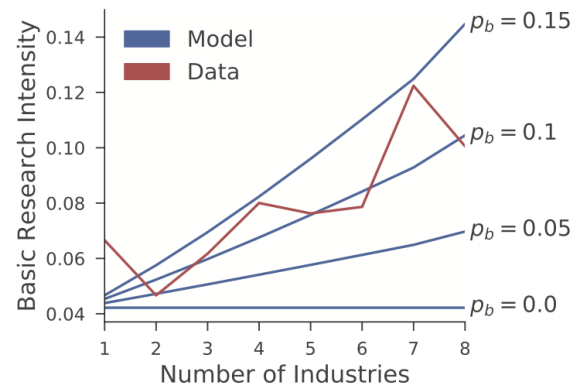


FIGURE 10

Basic research intensity versus spillover parameter p

5.6 Tables 5–6 and Figures 11–12: parameter estimates and fit

TABLE 5
Parameter estimates

#	Description	Sym	Value	#	Description	Sym	Value
1.	Discount rate	δ	0.039	10.	Applied cost curvature	v_a	1.367
2.	CRRA utility parameter	γ	3.020	11.	Basic cost curvature	v_b	1.539
3.	Elasticity of substitution	ε	5.863	12.	Applied cost scale	ξ_a	1.230
4.	Cross-industry spillover	p	0.116	13.	Basic cost scale	ξ_b	5.445
5.	Buyout rate	ς	0.456	14.	Exogenous exit rate	ε	0.006
6.	Basic step size	η	0.079	15.	Basic fixed mean	ϕ	-4.759
7.	Applied step size	λ	0.050	16.	Basic fixed Std. Dev.	σ	0.329
8.	Mass of entrants	E	0.495	17.	Product cooldown rate	ζ	0.117
9.	Mass of academic labs	U	0.493	18.	Citation rate	x	2.984

TABLE 6
Moments used in estimation

#	Description	Model	Data	#	Description	Model	Data
$\Lambda(1-8)$	Basic research extensive	See Figure 12		$\Lambda(23)$	R&D/labour	0.276	0.260
$\Lambda(9-16)$	Basic research intensive	See Figure 11		$\Lambda(24)$	Employment growth	0.103	0.103
$\Lambda(17)$	Mean industries	2.264	2.203	$\Lambda(25)$	Aggregate growth	0.013	0.015
$\Lambda(18)$	Mean square industries	7.535	6.976	$\Lambda(26)$	Spillover differential	8.344	8.000
$\Lambda(19)$	Return on sales	0.033	0.033	$\Lambda(27)$	Public citations mean	8.682	7.013
$\Lambda(20)$	Exit rate	0.092	0.092	$\Lambda(28)$	Public citations RMS	12.63	14.20
$\Lambda(21)$	Age, small firms	11.77	15.00	$\Lambda(29)$	Private citations mean	5.747	5.885
$\Lambda(22)$	Age, large firms	19.22	24.87	$\Lambda(30)$	Private citations RMS	8.581	9.154

TABLE 11
Optimal academic policies (in percentages)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc plan	—	—	—	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61
Uniform	29.7	29.7	0.5	16.7	1.5	0.10	82.1	1.5	0.49	5.3	10.7	13.4	1.68	100.64
Targeted	11.1	48.6	0.5	12.7	4.4	0.11	83.5	5.1	0.50	3.5	7.3	29.9	1.71	102.81
Unif Acad	29.8	29.8	0.68	16.7	1.46	0.27	81.9	1.47	0.67	5.3	10.7	15.4	1.70	100.70
Unif Acad+	24.3	24.3	3.24	15.5	0.89	2.80	81.7	0.86	3.19	4.7	9.6	24.4	1.88	102.88

See Table 7 for variable definitions.

TABLE 12
Optimal subsidies with various academic productivities

Academic multiplier	Optimal uniform subsidy	Optimal academic spending	Welfare gain
1x	29.8	0.68	0.7
2x	29.2	1.01	1.0
5x	27.6	1.54	1.9

Read:

- Table 5 reports the core structural estimates: $p = 0.116$, $\eta = 0.079$, $\lambda = 0.050$, and $\zeta = 0.117$.
- Table 6 shows the model matches the targeted moments reasonably well, including industry scope, profitability, growth, and citation moments.
- Figures 11 and 12 compare model versus data for the extensive and intensive margins of basic research by industry scope. Both fit the central empirical pattern well.

5.7 Tables 7–10: baseline, planner, and subsidy policy

TABLE 7
Decentralized economy: endogenous variables (in percentages)

ψ	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Subsidy rate	Public basic innovation in GDP	Applied basic innovation	Private basic innovation	Public basic innovation	Production labour	Labour in private basic	Labour in public basic	Entrant labour	Labour in private applied	Share of hot product lines	Growth rate	Welfare
10	0.5	14.7	0.60	0.13	85.6	0.55	0.52	4.4	8.9	7.2	1.35	100.0

TABLE 9
Uniform research subsidy (in percentages)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc plan	—	—	—	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61
Uniform	29.7	29.7	0.5	16.7	1.5	0.10	82.1	1.5	0.49	5.3	10.7	13.4	1.68	100.64

See Table 7 for variable definitions.

TABLE 8
Social planner's optimum (in percentages)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc plan	—	—	—	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61

See Table 7 for variable definitions.

TABLE 10
Type-dependent research subsidy (in percentages)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ_b^e	τ_u	L_p	L_b	L_u	L_e	L_a	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc plan	—	—	—	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61
Uniform	29.7	29.7	0.5	16.7	1.5	0.10	82.1	1.5	0.49	5.3	10.7	13.4	1.68	100.64
Targeted	11.1	48.6	0.5	12.7	4.4	0.11	83.5	5.1	0.50	3.5	7.3	29.9	1.71	102.81

See Table 7 for variable definitions.

Read:

- Table 7 describes the decentralized economy: only about 7% of product lines are hot, and only a small share of research labor goes to basic research.
- Table 8 shows the planner shifts strongly toward basic research: the hot share rises from 7.2% to 35.4%, growth rises from 1.35% to 1.88%, and welfare rises by 4.61% consumption-equivalent.
- Table 9 shows why the uniform subsidy disappoints: it raises total research but pushes too much effort into applied research. Welfare improves only from 100.00 to 100.64.
- Table 10 is the headline policy table: the optimal targeted subsidy is about 11.1% for applied research and 48.6% for basic research, with welfare rising to 102.81.

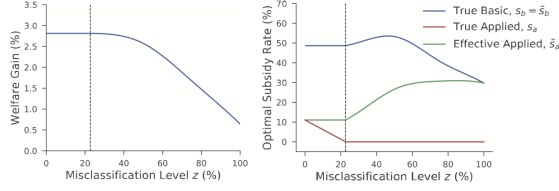


FIGURE 13
Optimal policies and welfare with misclassification of research

5.8 Figure 13 and Tables 11–12: implementation and academic policy

Read:

- Figure 13 shows that targeted-subsidy gains are robust to substantial misclassification. Welfare stays close to its targeted-policy value until misclassification becomes very large.
- Table 11 shows that optimal academic spending rises relative to baseline, and that public basic research complements rather than substitutes for private research support.
- Table 12 shows that when academic labs are more productive, the optimal public-research budget increases and welfare gains become much larger.

TABLE 11
Optimal academic policies (in percentages)

Policy	ψ_a	ψ_b	R/Z	τ_a	τ'_a	τ_b	L_p	L_b	L_a	L_e	L_u	α	g	β
Baseline	10.0	10.0	0.5	14.7	0.6	0.13	85.6	0.6	0.52	4.4	8.9	7.2	1.35	100.00
Soc plan	—	—	—	12.0	3.8	1.89	83.1	4.5	2.22	3.4	6.8	35.4	1.88	104.61
Uniform	29.7	29.7	0.5	16.7	1.5	0.10	82.1	1.5	0.49	5.3	10.7	13.4	1.68	100.64
Targeted	11.1	48.6	0.5	12.7	4.4	0.11	83.5	5.1	0.50	3.5	7.3	29.9	1.71	102.81
Unif Acad	29.8	29.8	0.68	16.7	1.46	0.27	81.9	1.47	0.67	5.3	10.7	15.4	1.70	100.70
Unif Acad+	24.3	24.3	3.24	15.5	0.89	2.80	81.7	0.86	3.19	4.7	9.6	24.4	1.88	102.88

See Table 7 for variable definitions.

TABLE 12
Optimal subsidies with various academic productivities

Academic multiplier	Optimal uniform subsidy	Optimal academic spending	Welfare gain
1×	29.8	0.68	0.7
2×	29.2	1.01	1.0
5×	27.6	1.54	1.9

6 Short summary

- **Conclusion.** Basic research matters because it generates both broad cross-industry spillovers and temporary within-industry opportunities for more productive applied innovation.
- **Intuition.** Private firms underinvest in basic research because they cannot appropriate enough of the spillovers, while competitive forces can push them to overinvest in applied research.
- **Empirical lesson.** Research composition, not just total R&D, is the right object for policy. Firm scope and citation dynamics are informative enough to discipline a structural model of this distinction.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The decentralized equilibrium suffers from dynamic misallocation in research composition: too little basic research and too much applied research.
- **Magnitude.** The paper estimates that the resulting welfare loss is about 4.6% in consumption-equivalent terms.

- **Why the result is believable.** The paper combines a clear reduced-form fact (basic-research intensity rises with industry scope), an IV design using historical state ownership, and structural discipline from patent citations and firm dynamics.
- **Mechanism evidence.** Public patents receive more citations and produce more valuable follow-on innovations for a while, which supports the “hot-line” mechanism of basic research.
- **Policy implication.** Uniform R&D subsidies are a blunt instrument because they accentuate the existing overinvestment in applied research. Better policy is to target basic research directly and to complement private incentives with public basic-research funding.

7.2 Economic intuition

- Basic research is socially valuable because its discoveries travel broadly, but firms only capture the subset that lands in industries where they already operate.
- Applied research is privately attractive because it turns existing knowledge into commercial products quickly; once the stock of basic knowledge rises, the private return to applied research rises too.
- This complementarity means that a policy raising basic research can increase the productivity of applied research even if it reduces applied-research subsidies.

7.3 Takeaway

This paper’s big lesson is that innovation policy should be designed around *knowledge spillovers*, not just around total spending. In the authors’ framework, the central margin is the composition of research between basic and applied work, together with the public–private division in who creates that knowledge. Once that distinction is taken seriously, the case for heavily supporting basic research becomes much stronger than standard uniform-subsidy logic would suggest.